JEE (Main) QUESTION PAPER

2025
29th Jan (Shift 2)

Time : 3 hours **Total Marks : 300**

General Instructions :

1. *There are three subjects in the question paper consisting of Physics (Q. no. 1 to 25), Chemistry (Q. no. 26 to 50) and Mathematics (Q. no. 51 to 75).*
2. *Each subject is divided into two sections. Section A consists of 20 multiple choice questions & Section B consists of 5 numerical value type questions. In Section B, candidates have to attempt any five questions out of 5.*
3. *There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice for Section A questions and zero mark will be awarded for not attempted question.*
4. *For Section B questions, 4 marks will be awarded for correct answer and zero for unattempted and incorrect answer.*
5. *Any textual, printed or written material, mobile phones, calculator, etc. is not allowed for the students appearing for the test.*
6. *All calculations / written work should be done in the rough sheet is provided with Question Paper.*

Physics

Section A

Q. 1. Difference in temperature in a material can convert heat energy into electrical energy. To harvest the heat energy, the material should have:

(1) low thermal conductivity and low electrical conductivity.
(2) high thermal conductivity and high electrical conductivity.
(3) low thermal conductivity and high electrical conductivity.
(4) high thermal conductivity and low electrical conductivity.

Q. 2. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): With the increase in pressure of an ideal gas, volume falls more rapidly in isothermal process than in adiabatic process.

Reason (R): In isothermal process, PV = constant, while in adiabatic process PV^{γ} = constant. Here γ is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas.

In light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(2) (A) is true but (R) is false.
(3) Both (A) and (R) are true and (R) is the correct explanation of (A).
(4) (A) is false but (R) is true.

Q. 3. An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_0. Choose the correct option from the following:

(1) Torque on the dipole is zero and the net force is directed away from the sheet.
(2) The torque on the dipole is zero and the net force acts towards the sheet.
(3) Potential energy of the dipole is minimum and the torque is zero.
(4) Potential energy and torque both are maximum.

Q. 4. In an experiment with the photoelectric effect, the stopping potential:

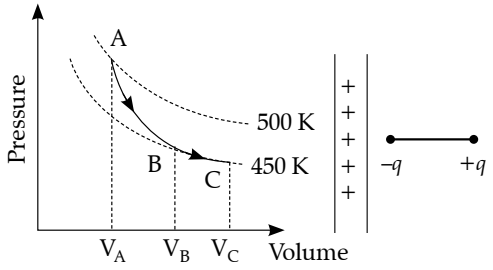
(1) increases with the increase in the wavelength of the incident light.
(2) increases with the increase in the intensity of the incident light.
(3) is $\left(\frac{1}{e}\right)$ times the maximum kinetic energy of the emitted photoelectrons.
(4) decreases with the increase in the intensity of the incident light.

Q. 5. A point charge causes an electric flux of $-2 \times 10^4\ Nm^2C^{-1}$ to pass through a spherical Gaussian surface with the radius of 8.0 cm centred on the charge.

The value of the point charge is:
(Given, $\varepsilon_0 = 8.85 \times 10^{-12} C^2N^{-1}m^{-2}$)

(1) -17.7×10^{-8}C **(2)** -15.7×10^{-8}C
(3) 17.7×10^{-8}C **(4)** 15.7×10^{-8}C

Q. 6.

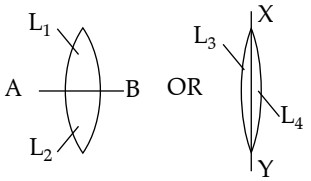


A polyatomic molecule (C_V = 3 R, C_P = 4 R, where R is gas constant) goes from phase space point A($P_A = 10^5$ Pa, $V_A = 4 \times 10^{-6} m^3$) to point B($P_B = 5 \times 10^4$ Pa, $V_B = 6 \times 10^{-6} m^3$) to point C($P_C = 10^4$ Pa, $V_C = 8 \times 10^{-6} m^3$). A to B is an adiabatic path and B to C is an isothermal path.

The net heat absorbed per unit mole by the system is:

(1) 500R(ln3 + ln4) **(2)** 450R(ln4 – ln3)
(3) 500R ln2 **(4)** 400R ln4

Q. 7. Two identical, symmetric double convex lenses of focal length *f* are cut into two equal parts L_1, L_2 by AB plane and L_3, L_4 by XY plane as shown in figure respectively. The ratio of focal lengths of lenses L_1 and L_3 is:

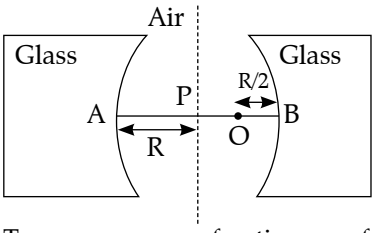


(1) 1 : 4 **(2)** 1 : 1
(3) 2 : 1 **(4)** 1 : 2

Q. 8. A plane electromagnetic wave propagates along the $+x$ direction in free space. The components of the electric field $\vec{E}$ and the magnetic field $\vec{B}$ vectors associated with the wave in the Cartesian frame are:

(1) E_y, B_x **(2)** E_y, B_z
(3) E_x, B_y **(4)** E_z, B_y

Q. 9.



Two concave refracting surfaces having equal radii of curvature and refractive index 1.5 face each other in air as shown in the figure above. A point object O is placed midway between P and B. The separation between the images of O formed by each refracting surface is:

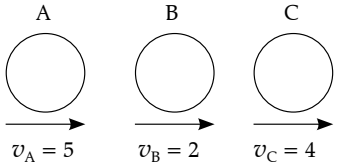
(1) 0.214 R **(2)** 0.114 R
(3) 0.411 R **(4)** 0.124 R

Q. 10. Two bodies, A and B, of equal mass are suspended from two massless springs of spring constant k_1 and k_2, respectively. If the bodies oscillate vertically such that their amplitudes are equal, the ratio of the maximum velocity of A to the maximum velocity of B is:

(1) $\sqrt{\frac{k_1}{k_2}}$ **(2)** $\frac{k_1}{k_2}$ **(3)** $\frac{k_2}{k_1}$ **(4)** $\sqrt{\frac{k_2}{k_1}}$

Q. 11. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):



Three identical spheres of the same mass undergo one dimensional motion as shown in the figure, with initial velocities $v_A = 5$ m/s, $v_B = 2$ m/s, $v_C = 4$ m/s. If we wait sufficiently long for elastic collision to happen, then $v_A = 4$ m/s, $v_B = 2$ m/s, $v_C = 5$ m/s will be the final velocities.

Reason (R): In an elastic collision between identical masses, two objects exchange their velocities.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(2) (A) is true but (R) is false.
(3) Both (A) and (R) are true and (R) is the correct explanation of (A).
(4) (A) is false but (R) is true.

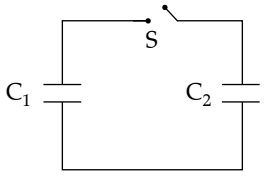
Q. 12. A sand dropper drops sand of mass $m(t)$ on a conveyer belt at a rate proportional to the square root of the speed (v) of the belt, i.e. $\frac{dm}{dt} \propto \sqrt{v}$. If P is the power delivered to run the belt at constant speed then which of the following relationships is true ?

(1) $P^2 \propto v^3$ **(2)** $P \propto \sqrt{v}$
(3) $P \propto v$ **(4)** $P^2 \propto v^5$

Q. 13. A convex lens made of glass (refractive index = 1.5) has focal length 24 cm in air. When it is totally immersed in water (refractive index = 1.33), its focal length changes to:

(1) 72 cm **(2)** 96 cm
(3) 24 cm **(4)** 48 cm

Q. 14. A capacitor, C_1 = 6 μF, is charged to a potential difference of V_0 = 5 V using a 5 V battery. The battery is removed and another capacitor, C_2 = 12 μF, is inserted in the place of the battery. When the switch 'S' is closed, the charge flows between the capacitors for some time until equilibrium condition is reached. What are the charges (q_1 and q_2) on the capacitors C_1 and C_2 when equilibrium condition is reached?



(1) $q_1 = 15\ \mu C, q_2 = 30\ \mu C$
(2) $q_1 = 30\ \mu C, q_2 = 15\ \mu C$
(3) $q_1 = 10\ \mu C, q_2 = 20\ \mu C$
(4) $q_1 = 20\ \mu C, q_2 = 10\ \mu C$

Q. 15. Three equal masses m are kept at vertices (A, B, C) of an equilateral triangle of side a in free space. At $t = 0$, they are given an initial velocity $\vec{V}_A = V_0\overrightarrow{AC}, \vec{V}_B = V_0\overrightarrow{BA}$ and $\vec{V}_C = V_0\overrightarrow{CA}$. Here, $\overrightarrow{AC}, \overrightarrow{CB}$ and $\overrightarrow{BA}$ are unit vectors along the edges of the triangle. If the three masses interact gravitationally, then the magnitude of the net angular momentum of the system at the point of collision is:

(1) $\frac{1}{2}amV_0$ **(2)** $3amV_0$
(3) $\frac{\sqrt{3}}{2}amV_0$ **(4)** $\frac{3}{2}amV_0$

Q. 16. Match List-I with List-II.

List-I	List-II
(A) Young's Modulus	**(I)** $ML^{-1}T^{-1}$
(B) Torque	**(II)** $ML^{-1}T^{-2}$
(C) Coefficient of Viscosity	**(III)** $M^{-1}L^{3}T^{-2}$
(D) Gravitational Constant	**(IV)** $ML^{2}T^{-2}$

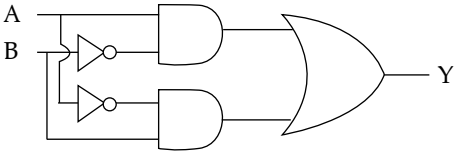
Choose the correct answer from the options given below:
(1) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
(2) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
(4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Q. 17. Match List-I with List-II.

List-I	List-II
(A) Magnetic induction	**(I)** ampere metre^2
(B) Magnetic intensity	**(II)** weber
(C) Magnetic flux	**(III)** gauss
(D) Magnetic moment	**(IV)** ampere/ metre

Choose the correct answer from the options given below:
(1) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
(2) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(3) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
(4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Q. 18. The truth table for the circuit is given below:



(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	0
1	0	0
1	1	0
0	1	1

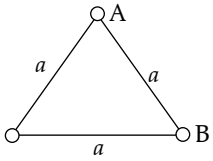
(3)

A	B	Y
0	0	0
1	0	1
0	1	0
1	1	0

(4)

A	B	Y
0	0	0
1	1	1
1	0	1
0	1	1

Q. 19. A cup of coffee cools from 90°C to 80°C in t minutes when the room temperature is 20°C. The time taken by a similar cup of coffee to cool from 80°C to 60°C at the same room temperature is:



(1) $\frac{13}{5}t$ **(2)** $\frac{10}{13}t$ **(3)** $\frac{13}{10}t$ **(4)** $\frac{5}{13}t$

Q. 20. The number of spectral lines emitted by atomic hydrogen that is in the 4th energy level is:
(1) 6 **(2)** 0 **(3)** 3 **(4)** 1

Section B

Q. 21. The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} tesla. If the solenoid carries a current of 0.29 A, then the length of the solenoid is _____ π cm.

Q. 22. A parallel plate capacitor consisting of two circular plates of radius 10 cm is being charged by a constant current of 0.15 A. If the rate of change of potential difference between the plates is 7×10^8 V/s then the integer value of the distance between the parallel plates is _____ μm.
$\left(\text{Take, } \varepsilon_0 = 9\times10^{-12}\frac{F}{m}, \pi = \frac{22}{7}\right)$

Q. 23. A physical quantity, Q, is related to four observables a, b, c, d as follows: $Q = \frac{ab^4}{cd}$ where, $a = (60 \pm 3)$Pa; $b = (20 \pm 0.1)$m; $c = (40 \pm 0.2)\ \text{Nsm}^{-2}$ and $d = (50 \pm 0.1)$m, then the percentage of error in Q is $\frac{x}{1000}$, where x = _____.

Q. 24. Two planets, A and B, are orbiting a common star in circular orbits of radii R_A and R_B, respectively, with $R_B = 2R_A$. The planet B is times more massive than planet A. The ratio $\left(\frac{L_B}{L_A}\right)$ of angular momentum (L_B) of planet B to that of planet A(L_A) is closest to integer _____.

Q. 25. Two cars, P and Q, are moving on a road in the same direction. Acceleration of car P increases linearly with time whereas car Q moves with a constant acceleration. Both cars cross each other at time $t = 0$ for the first time. The maximum possible number of crossing(s) (including the crossing at $t = 0$) is _____ .

Chemistry

Section A

Q. 26. The calculated spin-only magnetic moments of $K_3[Fe(OH)_6]$ and $K_4[Fe(OH)_6]$ respectively are:

(1) 4.90 and 4.90 B.M.

(2) 5.92 and 4.90 B.M.

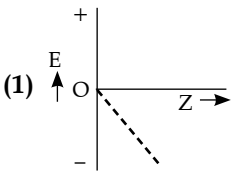
(3) 3.87 and 4.90 B.M.

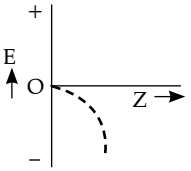
(4) 4.90 and 5.92 B.M.

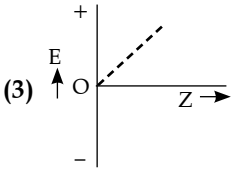
Q. 27. For hydrogen-like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n?

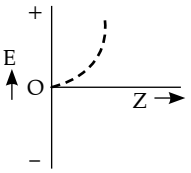
[E : Energy of the stationary state,

Z: atomic number, n = principal quantum number]

(1) 

(2) 

(3) 

(4) 

Q. 28. Given below are two statements:

Statement (I): In partition chromatography, stationary phase is the thin film of liquid present in the inert support.

Statement (II): In paper chromatography, the material of paper acts as a stationary phase.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are false.

(2) Statement I is true but Statement II is false.

(3) Both Statement I and Statement II are true.

(4) Statement I is false but Statement II is true.

Q. 29. Identify essential amino acids from those listed below:

(A) Valine **(B)** Proline

(C) Lysine **(D)** Threonine

(E) Tyrosine

Choose the correct answer from the options given below:

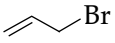
(1) (A), (C) and (D) only

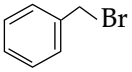
(2) (A), (C) and (E) only

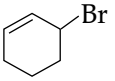
(3) (B), (C) and (E) only

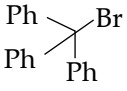
(4) (C), (D) and (E) only

Q. 30. Which among the following halides will generate the most stable carbocation in a nucleophillic substitution reaction?

(1) 

(2) 

(3) 

(4) 

Q. 31. Consider the equilibrium-

$CO(g) + 3H_2(g) \rightarrow CH_4(g) + H_2O(g)$

If the pressure applied over the system increases twofold at constant temperature then:

(A) The concentration of reactants and products increases.

(B) Equilibrium will shift in the forward direction.

(C) Equilibrium constant increases as the concentration of products increases.

(D) Equilibrium constant remains unchanged as the concentration of reactants and products remain the same.

Choose the correct answer from the options given below:

(1) (A) and (B) only

(2) (A), (B) and (D) only

(3) (B) and (C) only

(4) (A), (B) and (C) only

Q. 32. Given below are two statements:

Statement (I): NaCl is added to the ice present in the ice cream box at 0°C to prevent the melting of ice cream.

Statement (II): On addition of NaCl to the ice at 0°C, there is a depression in the freezing point.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true.

(2) Both Statement I and Statement II are true.

(3) Both Statement I and Statement II are false.

(4) Statement I is true but Statement II is false.

Q. 33. Given below are two statements:

Statement (I): On nitration of m-xylene with HNO_3 and H_2SO_4 followed by oxidation, 4-nitrobenzene-1 and 3-dicarboxylic acid are obtained as the major products.

Statement (II): The CH_3 group is a o/p-directing while the NO_2 group is a m-directing group.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Statement I is true but Statement II is false.

Q. 34. 0.1m solution of KI reacts with excess H_2SO_4 and KIO_3 solution. According to the equation

$5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

Identify the correct statements :

(A) 200 mL of KI solution reacts with 0.004 mol of KIO_3.

(B) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4.

(C) 0.5 L of KI solution produced 0.005 mol of I_2.

(D) Equivalent weight of KIO_3 is equal to $\left(\frac{\text{Molecular weight}}{5}\right)$.

Choose the correct answer from the options given below:

(1) (A) and (D) only **(2)** (B) and (C) only

(3) (A) and (B) only **(4)** (C) and (D) only

Q. 35. Match List-I with List-II:

List-I Applications	**List-II Batteries/Cell**
(A) Transistors	**(I)** Anode–Zn/Hg; Cathode–HgO + C
(B) Hearing aids	**(II)** Hydrogen fuel cell
(C) Invertors	**(III)** Anode – Zn; Cathode - Carbon
(D) Apollo space ship	**(IV)** Anode – Pb; Cathode – Pb | PbO_2

Choose the correct answer from the options given below:

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(2) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(4) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

Q. 36. O_2 gas will evolve as a product of the electrolysis of:

(A) an aqueous solution of $AgNO_3$ using silver electrodes.

(B) an aqueous solution of $AgNO_3$ using platinum electrodes.

(C) a diluted solution of H_2SO_4 using platinum electrodes.

(D) a high-concentration solution of H_2SO_4 using platinum electrodes.

Choose the correct answer from the options given below:

(1) (B) and (C) only **(2)** (A) and (D) only

(3) (B) and (D) only **(4)** (A) and (C) only

Q. 37. Identify homoleptic complexes with an odd number of *d* electrons in the central metal.

(A) $[FeO_4]^{2-}$ (B) $[Fe(CN)_6]^{3-}$

(C) $[Fe(CN)_5NO]^{2-}$ (D) $[CoCl_4]^{2-}$

(E) $[Co(H_2O)_3F_3]$

Choose the correct answer from the options given below:

(1) (B) and (D) only
(2) (C) and (E) only
(3) (A), (B) and (D) only
(4) (A), (C) and (E) only

Q. 38. The total number of sigma (σ) is_____ and pi(π) is____ bonds in hex-1-en-4-yne

(1) 13 and 3 **(2)** 11 and 3
(3) 3 and 13 **(4)** 14 and 3

Q. 39. If C(diamond) → C(graphite) + X kJ mol^{-1}
C(diamond) + $O_2(g)$ → $CO_2(g)$ + Y kJ mol^{-1}
C(graphite) + $O_2(g)$ → $CO_2(g)$ + Z kJ mol^{-1}
at constant temperature, then:

(1) X = Y + Z **(2)** –X = Y + Z
(3) X = –Y + Z **(4)** X = Y – Z

Q. 40. Given below are two statements:

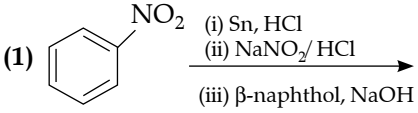
Statement (I): It is impossible to specify simultaneously, with arbitrary precision, both linear momentum and the position of a particle.

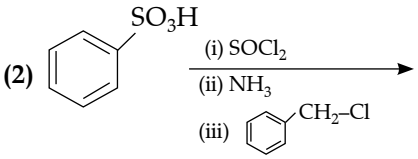
Statement (II): If the uncertainty in the measurement of position and the uncertainty in the measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\geq \sqrt{\frac{h}{\pi}} \times \frac{1}{2m}$

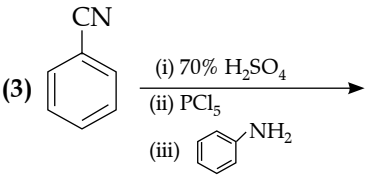
In the light of the above statements, choose the correct answer from the options given below:

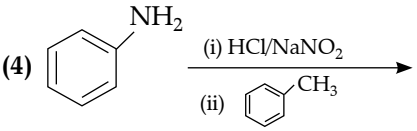
(1) Statement I is true but Statement II is false.
(2) Both Statement I and Statement II are true.
(3) Statement I is false but Statement II is true.
(4) Both Statement I and Statement II are false.

Q. 41. Which one of the following reaction sequences will give an azo dye?

(1) 

(2) 

(3) 

(4) 

Q. 42. Drug X becomes ineffective after 50% decomposition. The original concentration of the drug in a bottle was 16 mg/mL, which becomes 4 mg/mL in 12 months. The expiry time of the drug in months is _____.

Assume that the decomposition of the drug follows first order kinetics.

(1) 12 **(2)** 2
(3) 3 **(4)** 6

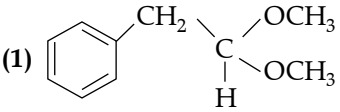
Q. 43. The type of oxide formed by the element amongst Li, Na, Be, Mg, B and Al, which has the least atomic radius is:

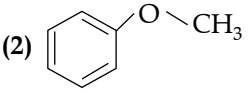
(1) A_2O_3 **(2)** AO_2
(3) AO **(4)** A_2O

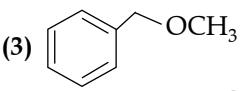
Q. 44. First ionisation enthalpy values of first four group 15 elements are given below. Choose the correct value for the element that is a main component of the apatite family:

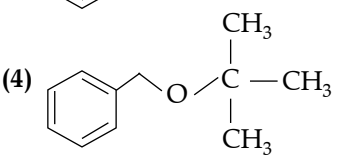
(1) 1012 kJ mol^{-1} **(2)** 1402 kJ mol^{-1}
(3) 834 kJ mol^{-1} (4) 947 kJ mol^{-1}

Q. 45. Which one of the following, will give a phenol with HBr?

(1) 

(2) 

(3) 

(4) 

Section B

Q. 46. Consider the following low-spin complexes: $K_3[Co(NO_2)_6]$, $K_4[Fe(CN)_6]$, $K_3[Fe(CN)_6]$, $Cu_2[Fe(CN)_6]$ and $Zn_2[Fe(CN)_6]$

The sum of the spin-only magnetic moment value of complexes having yellow colour is _____ B.M. (the answer is the nearest integer).

Q. 47. Isomeric hydrocarbons → negative Baeyer's test (Molecular formula C_9H_{12}).

The total number of isomers from above,with four different non-aliphatic substitution sites is.....

Q. 48. In the Claisen-Schmidt reaction, to prepare dibenzalacetone from 5.3 g benzaldehyde, a total of 3.51 g of product was obtained. The percentage yield in this reaction was ____ %.

Q. 49. In sulphur estimation, 0.20 g of a pure organic compound gave 0.40 g of barium sulphate. The percentage of sulphur in the compound is ____ $\times 10^{-1}$ %.

(Molar "mass": O = 16, S = 32,

Ba = 137 in g mol^{-1})

Q. 50. Total number of non-bonded electrons present in the NO_2^- ion based on the Lewis theory is _____ .

Mathematics

Section A

Q. 51. If the set of all $a \in R$, for which the equation $2x^2 + (a-5)x + 15 = 3a$ has no real root, is the interval (α, β), and $X = \{x \in Z : \alpha < x < \beta\}$, then $\sum_{x \in X} x^2$ equal to:

(1) 2109 **(2)** 2129 **(3)** 2139 **(4)** 2119

Q. 52. If $\sin x + \sin^2 x = 1, x \in \left(0, \frac{\pi}{2}\right)$, then $(\cos^{12}x + \tan^{12}x) + 3(\cos^{10}x + \tan^{10}x + \cos^8 x + \tan^8 x) + (\cos^6 x + \tan^6 x)$ is equal to:

(1) 4 **(2)** 3 **(3)** 2 **(4)** 1

Q. 53. Let the area enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ be α. If $9\alpha = \beta\pi + \gamma$; β, γ are integers, then the value of $|\beta - \gamma|$ equals:

(1) 27 **(2)** 18 **(3)** 15 **(4)** 33

Q. 54. If the domain of the function $\log_5(18x - x^2 - 77)$ is (α, β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ, δ), then $(\alpha^2 + \beta^2 + \gamma^2)$ is equal to:

(1) 195 **(2)** 174 **(3)** 186 **(4)** 179

Q. 55. Let the function $f(x) = (x^2-1)|x^2 - ax + 2| + \cos|x|$ be not differentiable at the two points $x = \alpha = 2$ and $x = \beta$. Then the distance of the point (α, β) from the line $12x + 5y + 10 = 0$ is equal to:

(1) 3 **(2)** 1 **(3)** 2 **(4)** 5

Q. 56. Let a straight line L pass through the point P(2, –1, 3) and be perpendicular to the lines $\frac{x-1}{2} = \frac{y+1}{1} = \frac{z-3}{-2}$ and $\frac{x-3}{1} = \frac{y-2}{3} = \frac{z+2}{4}$

If the line L intersects the yz-plane at the point Q, then the distance between the points P and Q is :

(1) 2 **(2)** $\sqrt{10}$ **(3)** 3 **(4)** $2\sqrt{3}$

Q. 57. Let $S = N \cup \{0\}$. Define a relation R from S to R by :

$$R = \left\{(x, y) : \log_e y = x \log_e\left(\frac{2}{5}\right), x \in S, y \in R\right\}.$$

Then, the sum of all the elements in the range of R is equal to:

(1) $\frac{3}{2}$ **(2)** $\frac{5}{3}$ **(3)** $\frac{10}{9}$ **(4)** $\frac{5}{2}$

Q. 58. Let the line $x + y = 1$ meet the axes of x and y at A and B, respectively. A right-angled triangle AMN is inscribed in the triangle OAB, where O is the origin and the points M and N lie on the lines OB and AB, respectively. If the area of the triangle AMN is $\frac{4}{9}$ of the area of the triangle OAB and AN : NB = λ : 1, then the sum of all possible value(s) of λ is :

(1) $\frac{1}{2}$ **(2)** $\frac{13}{6}$ **(3)** $\frac{5}{2}$ **(4)** 2

Q. 59. If $\alpha x + \beta y = 109$ is the equation of the chord of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, whose mid point is $\left(\frac{5}{2}, \frac{1}{2}\right)$, then $\alpha + \beta$ is equal to:

(1) 37 **(2)** 46 **(3)** 58 (4) 72

Q. 60. If all the words with or without meaning made using all the letters of the word "KANPUR" are arranged as in a dictionary, then the word at 440th position in this arrangement, is :

(1) PRNAKU **(2)** PRKANU
(3) PRKAUN **(4)** PRNAUK

Q. 61. Let α, β ($\alpha \neq \beta$) be the values of m, for which the equations $x + y + z = 1$; $x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10}(n^{\alpha} + n^{\beta})$ is equal to :

(1) 440 **(2)** 3080 **(3)** 3410 **(4)** 560

Q. 62. Let $A = [a_{ij}]$ be a matrix of order 3×3, with $a_{ij} = \left(\sqrt{2}\right)^{i+j}$. If the sum of all the elements in the third row of A^2 is $\alpha + \beta\sqrt{2}$, $\alpha, \beta \in Z$, then $\alpha + \beta$ is equal to:

(1) 280 **(2)** 168 **(3)** 210 (4) 224

Q. 63. Let P be the foot of the perpendicular from the point (1, 2, 2) on the line $L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$.
Let the line $\vec{r} = (-\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$, $\lambda \in$ R, intersect the line L at Q. Then $2(PQ)^2$ is equal to:

(1) 27 **(2)** 25 **(3)** 29 **(4)** 19

Q. 64. Let a circle C pass through the points (4, 2) and (0, 2), and its centre lie on $3x + 2y + 2 = 0$ Then the length of the chord, of the circle C, whose midpoint is (1, 2), is:

(1) $\sqrt{3}$ **(2)** $2\sqrt{3}$ **(3)** $4\sqrt{2}$ **(4)** $2\sqrt{2}$

Q. 65. Let $A = [a_{ij}]$ be a 2×2 matrix such that $a_{ij} \in \{0, 1\}$ for all i and j. Let the random variable X denote the possible values of the determinant of the matrix A. Then, the variance of X is:

(1) $\frac{1}{4}$ **(2)** $\frac{3}{8}$ **(3)** $\frac{5}{8}$ **(4)** $\frac{3}{4}$

Q. 66. Bag 1 contains 4 white balls and 5 black balls, and Bag 2 contains n white balls and 3 black balls. One ball is drawn randomly from Bag 1 and transferred to Bag 2. A ball is then drawn randomly from Bag 2. If the probability, that the ball drawn is white, is 29/45, then n is equal to:

(1) 3 **(2)** 4 **(3)** 5 **(4)** 6

Q. 67. The remainder, when 7^{103} is divided by 23, is equal to:

(1) 14 **(2)** 9 **(3)** 17 **(4)** 6

Q. 68. Let $f(x) = \int_0^x t(t^2 - 9t + 20)dt$, $1 \le x \le 5$. If the range of f is $[\alpha, \beta]$, then $4(\alpha + \beta)$ equals:

(1) 157 **(2)** 253 **(3)** 125 **(4)** 154

Q 69. Let $\hat{a}$ be a unit vector perpendicular to the vectors $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} - \hat{k}$, and make an angle of $\cos^{-1}\left(-\frac{1}{3}\right)$ with the vector $\hat{i} + \hat{j} + \hat{k}$. If $\hat{a}$ makes an angle of $-\frac{\pi}{3}$ with the vector $\hat{i} + \alpha\hat{j} + \hat{k}$, then the value of α is :

(1) $-\sqrt{3}$ **(2)** $\sqrt{6}$ **(3)** $-\sqrt{6}$ **(4)** $\sqrt{3}$

Q. 70. If for the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + (\tan x)y = \frac{2 + \sec x}{(1 + 2\sec x)^2}$, $x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$, $f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{10}$, then $f\left(\frac{\pi}{4}\right)$ is equal to:

(1) $\frac{9\sqrt{3}+3}{10(4+\sqrt{3})}$ **(2)** $\frac{\sqrt{3}+1}{10(4+\sqrt{3})}$

(3) $\frac{5-\sqrt{3}}{2\sqrt{2}}$ **(4)** $\frac{4-\sqrt{2}}{14}$

Section B

Q. 71. If $24\int_0^{\pi/4}\left(\sin\left|4x - \frac{\pi}{12}\right| + [2\sin x]\right)dx = 2\pi + \alpha$, where [.] denotes the greatest integer function, then α is equal to _____.

Q. 72. If $\lim_{t\to 0}\left(\int_0^1 (3x+5)^t dx\right)^{\frac{1}{t}} = \frac{\alpha}{5e}\left(\frac{8}{5}\right)^{2/3}$, then α is equal to _____.

Q. 73. Let $a_1, a_2, \ldots, a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} + a_{15} + \ldots + a_{2020}) + a_{2024} = 2233$. Then $a_1 + a_2 + a_3 + \ldots + a_{2024}$ is equal to _____.

Q. 74. Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b), for which $\left|\frac{z-a}{z+b}\right| = 1$ and $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1$, $z \in C$, where ω and ω^2 are the roots of $x^2 + x + 1 = 0$, is equal to _____.

Q. 75. Let $y^2 = 12x$ the parabola and S be its focus. Let PQ be a focal chord of the parabola such that (SP) (SQ) $= \frac{147}{4}$. Let C be the circle described taking PQ as a diametre. If the equation of a circle C is $64x^2 + 64y^2 - \alpha x - 64\sqrt{3}y = \beta$, then $\beta - \alpha$ is equal to _____.

■■■